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Are middlemen facilitators or barriers to improve smallholders' welfare in rural economies? Empirical evidence from Ethiopia

Gumataw K. Abebe [a, *], Jos Bijman [b], Annie Royer [c]

[a] Department of Agricultural Sciences, American University of Beirut, Beirut, Lebanon
[b] Management Studies Group, Wageningen University, The Netherlands
[c] Laval University, Quebec City, Canada

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ABSTRACT

Middlemen play an important role by linking farmers to traders and final markets. This is particularly the case in developing countries, where market failure is ubiquitous and food chains still consist of many stages. This paper analyzes the factors affecting the farmers' decision to trade through middlemen and the impact of this choice on income and commercialization. We conceptualize middlemen both as an economic institution and as a social network structure. On the basis of data from 345 farmers in Ethiopia we found that several socioeconomic variables – particularly age, education, farm size, wealth and location – and social network variables – notably ethnic and religious ties – have an influence on farmers' choice of sales arrangement. Regarding income effects, gross profit was 225% higher for farmers without intermediation. This could be explained by the latter farmers having access to better quality inputs, better contract specifications and receiving higher prices for their products. Nonetheless, the majority of farmers continue trading via middlemen. We suggest three explanations for this outcome. First, wholesalers seem to prefer to work with middlemen to guarantee minimum quantity and quality, and to reduce the cost of measuring quality. Second, personalized relationships might lock-in smallholders into trading through middlemen regardless of income losses. Third, trading via middlemen can enhance smallholder commercialization by linking low resource endowed farmers to traders and final markets. However, direct trading with wholesalers seemed beneficial for relatively better-resource endowed farmers.

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1. Introduction

Food value chains in developing economies are experiencing structural changes due to the rapid expansion of supermarkets and the increasingly strict quality and safety requirements of domestic and foreign customers (Henson and Jaffee, 2008; Swinnen and Maertens, 2007). These changes represent challenges as well as opportunities for smallholder farmers. All around the world, smallholders are striving to comply with these increasingly stringent requirements. Whether they are successful depends to a large extent on the way production and distribution is coordinated along the food chain (Hernández et al., 2007; Neven et al., 2009). Contract farming schemes, vertical integration and producer organizations are cited as important institutions that have emerged to address the market access challenges (Bijman, 2008; Henson et al., 2005; Moustier et al., 2010). Still, middlemen continue to play an important role in linking smallholder farmers to traders and final markets.

Although middlemen are active in developing economies' agricultural chains, their effect on improving the economic outcomes of smallholders is not clear, and they may have undesirable welfare[1] consequences; for instance, due to the lock-in effects farmers may not benefit from new market opportunities. Prior research on economic development has largely focused on individuals and institutions such as markets, firms, governments, and households but has paid less attention to personal relationships (Fafchamps, 2006).

We conceptualize middlemen not just as an economic institution that facilitates trade, but also as a social network structure.

[1] The conceptualization of welfare in this paper only focuses on economic performance - income and enhanced commercialization. In the general literature, welfare could be captured by the level of income, asset, or amount of expenditure (Sahn and Stifel, 2003).

* Corresponding author.
E-mail address: gumataw@gmail.com (G.K. Abebe).

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More specifically, we explore the social relationships among farmers and middlemen and analyze the impact of these relationships on the decision of the farmer to trade through a middleman (where this decision has welfare implications). Thus, our study connects with the growing body of literature on the impact of social networks on (farmer) choices and thereby on economic outcomes (e.g. Herforth et al., 2015; Michelson, 2015; World Bank, 2015).

We study the role of middlemen in Ethiopia where agriculture employs more than 80% of the population and farmers are embedded in several social network structures. Some of these include labor sharing networks ('debbo', 'wenfel'), informal insurance groups ('iddir'), rotating savings and credit associations ('equb'), as well as various kinds of oxen-sharing and crop-sharing arrangements (Krishnan and Sciubba, 2009). However, the effect of these networks on welfare has not received much attention in the empirical literature. Notable exceptions are Krishnan and Sciubba (2009) and Gabre-Madhin (2001). Using a network formation model and data from a household survey conducted in 1994 in rural Ethiopia, Krishnan and Sciubba (2009) illustrate the impact of labor sharing networks on economic performance of farmers. Gabre-Madhin (2001), in her study on intermediation in Ethiopian grain markets, found that middlemen constitute a socially optimal choice in economizing on search costs. The Ethiopian grain market is characterized by a lack of information on market prices, the product quality and quantity of supply. Traders consider middlemen as an important institution to avoid commitment failures. Gabre-Madhin (2001) found that middlemen in the grain market operate on a commission basis and transact on behalf of traders. In representing traders, they facilitate the supply of grain from different regions and they set daily prices through rapid appraisal of supply and demand conditions. According to Gabre-Madhin (2001), middlemen are particularly beneficial for traders.

Our paper seeks to extend the work by Gabre-Madhin (2001) by exploring the effect of middlemen on smallholders' welfare. For this purpose, we have conceptualized the community of farmers and middlemen as a social network structure. Our study deals with middlemen who act as facilitators of trade between farmers and wholesalers. The main objectives of our paper are the following: (1) to investigate the determinants of the choice of farmers for trading through middlemen; and (2) to analyze the effect of choosing a middleman or a wholesaler as their trading partner on the welfare of farmers. To achieve these objectives, we collected data through a survey in 2011 among 345 farmers in the Rift Valley region of Ethiopia. We focused on farmers who were producing potato (among other crops) because middlemen are an important institution in the potato market.

The paper is structured as follows. In Section 2, we provide the theoretical and methodological framework. Section 3 details the methods used to answer our research questions. Section 4 presents the results and discussion. Finally, Section 5 provides conclusion and suggestions for further research.

2. Theoretical and methodological framework

2.1. Theoretical framework

In economic theory, the main function of institutions is to minimize transaction costs (North, 1994; Platteau, 1994; Williamson, 1998). The transaction cost literature considers the institution of middlemen as a cost-reducing arrangement between sellers and buyers (Landa, 1981). Middlemen can economize on direct exchange costs, for example, by facilitating a match between a buyer and a seller (Biglaiser and Friedman, 1999) and by decreasing transaction costs related to search time and information asymmetry (Dixit, 2009; Hayami, 1996). Using a game theoretic approach, Townsend (1978) demonstrates that intermediaries emerge endogenously to economize on the fixed cost of exchange. Rubinstein and Wolinsky (1987) argue that middlemen are a time-saving institution since they shorten the negotiation time of sellers and buyers in a transaction.

Introducing the notion of quality uncertainty in the sense of Akerlof (1970), Li (1998) argues that middlemen provide efficiency benefits due to their role in bridging information asymmetry. Looking at the severity of the private information problem and the cost of the middleman's quality-testing technology, Li explains that if the information problem is not severe and if agents are willing to undertake exchange without knowing the exact quality of the goods, the presence of middlemen in trade is inefficient. However, if the information asymmetry prevents agents from trading since they cannot assess the quality of the goods, then middlemen can be welfare-improving. Therefore, middlemen are an efficient institution in markets where quality uncertainty is high. In contrast, Masters (2008) argues that the most persuasive and the least productive individuals are the ones who become middlemen to take advantage of the existence of producers who have lower production costs than themselves, but do not know the market. Under such market conditions, middlemen prosper by 'buying low' and 'selling high' and, hence, middlemen represent the cost of reducing information asymmetry.

The institution of middlemen differs from other types of economic institutions because it could also be considered as a social network structure, defined by group-membership (Munshi, 2014) and institutionalization of group relations (Portes, 1998). Wasserman and Faust (1994:20) define a social network as "a finite set or sets of actors and the relation or relations defined on them".[2] Thus, an economic transaction involving a middleman can represent a socially-tied relationship. According to Munshi (2014), close-knit communities such as those based on kinship and geographical proximity are characterized by strong social ties. Social networks become more important under conditions of contract uncertainty and positive transaction costs (Landa, 1981). They can prevent commitment failures, particularly in imperfect markets, because of their threat of social sanctions (Karlan et al., 2009), promote social capital (Coleman, 1988) and allow members to work together to achieve common objectives (Munshi, 2014). Karlan et al. (2009) argue that social networks can create trust and provide assurance in informal transactions in the same way collateral works in formal transactions. Two types of trust can develop in social networks – personalized and generalized trust. Personalized trust arises from repeated interpersonal interactions (Fafchamps, 2006), while generalized trust results from knowing the community or society as a whole (Platteau, 1994).

A drawback with a social network, however, is that it can force individuals to cooperate according to social norms even if this is against their self-interest (Fehr et al., 1997; Hoffman et al., 1998). It may decrease efficiency, for instance, by excluding new entrants from participating in the market, by applying price collusion and by constraining entrepreneurship (McMillan and Woodruff, 1999; Fafchamps, 2000). In addition, social networks may lead to lock-in effects (Jones et al., 2007).

2.2. Methodological framework

For the purpose of this study, we define the institution of middlemen as a special type of social network structure that is embedded in a community and facilitates economic exchange

[2] According to the same authors a relation is used to refer to the collection of ties of a given kind.

Table 1
Variable definition and descriptive statistics of sampled farmers.

Independent variables	Farmers selling through middlemen (n = 235)		Farmers selling directly to wholesalers (n = 110)	
	Mean	Std. dev.	Mean	Std. dev.
Age of the respondent (in years)	37.15	10.62	35.97	10.40
Gender (% male)	98	–	94	–
Family size	9.69	5.19	9.32	5.12
Education in school years (respondent)	5.77	3.59	5.42	3.44
Total livestock units***	5.0	5.09	14.67	28.36
Presence of a mobile phone (1 = %yes)	69	–	63	–
Presence of a cart (1 = %yes)	40	–	44	–
Potato land in 2005/6 season (ha)***	1.02	–	1.33	–
Total land size 2008/9 (ha)**	2.05	1.23	2.43	2.37
Membership in a Cooperative (%yes)	58	–	50	–
Household head has ethnic ties with the main middleman or buyer (%yes)	74	–	31	–
Household head has religious ties with the main middleman or buyer (% yes)	67	–	25	–
Cash received from potato sales 2009/10 (Birr)***	7581	596	13,368	1977
Income (cash received) from other crop sales 2008/9 season (Birr[a])***	6832	9681	9226	15,648
Income (cash received) from other sources, including from sale of cattle 2008/9 season (Birr)**	2172	3519	3441	8438
Farmer is from Shashemene district (%yes)	63	–	72	–
Farmer is from Shiraro district (%yes)	26	–	12	–
Farmer is from Shala district (%yes)	11	–	16	–

[a] 1US$ was approximately equal to 12.60 Birr in 2009/2010 season. *p < 0.1; **p < 0.05; ***p < 0.01.

between individuals from that community and outside traders. Smallholders' decisions to sell directly to wholesalers or to sell through a middleman (who intermediates between farmer and wholesaler) can be conceptualized as a binary channel choice decision problem by farmers that try to maximize utility. Most of the empirical literature on smallholders' channel choice emphasizes the effect of transaction costs and socioeconomic factors such as risk preferences, asset holdings, and geography (Alene et al., 2008; Barrett, 2008; Fischer and Qaim, 2012). Our study adds to this literature by including variables related to social network (ethnic and religious ties), human capital (age and education level), access to information, (access to a mobile phone), access to inputs (membership in a cooperative), access to market (the location of the farm relative to the location of the market), resource endowment (labor, cart, and number of livestock units), and volume of production (measured by land size).

Summary statistics for the above listed variables are presented in Table 1. Age and education level are often used as proxies for human capital (Fischer and Qaim, 2012). We expect that the probability of using a middleman diminishes with farmers' age and education level, because older and better educated farmers tend to have more experience and skills to assess the demands of wholesalers (Monson et al., 2008). Middlemen are generally considered important providers of market information. However, access to a mobile phone may reduce the need for other information providers (Muto and Yamano, 2009). We thus expect that a farmer who owns a mobile phone is less dependent on middlemen. Likewise, membership in a cooperative can provide better access to inputs (Kassie et al., 2011); thus, we expect members of a cooperative to make less use of middlemen.

Social networks play a significant role in trading relations in Ethiopia (Ali and Peerlings, 2011; Gabre-Madhin, 2001). In social networks, social norms of behavior are important determinants of human behavior, particularly in rural communities. We thus expect a strong relationship between our social network variables (ethnic and religious ties) and farmers' choice for the middleman as trading partner.

We used location dummies as a proxy for access to market (Hagos et al., 2012). Accordingly, farmers who are closer to a market are less likely to rely on middlemen since they are better positioned to find a buyer as they are more likely to be visited by one or more wholesalers. Also volume of production can influence the choice of a trading partner. Farmers who have high production volume are more likely to use intermediation than those farmers with low production volume because of the higher market risk the larger farmers face. However, it could also be argued that high volume creates economies of scale and thus may attract wholesalers. We used potato land size as a proxy for volume. In terms of resource endowment, farmers who own at least a cart are more likely to sell directly to wholesalers because they can easily transport their potatoes to the nearest market. Likewise, a large family size can provide more labor to carry potatoes to the nearest market in case no buyer comes to the farm gate (LeRoux et al., 2010). Farmers with a large number of livestock units are more likely to use middlemen because of the high opportunity cost of labor in searching for a buyer (Fafchamps and Hill, 2005).

3. Method

3.1. Study context and data

3.1.1. Study context

Potato is considered one of the spearheads of agricultural policy by the Ethiopian government because of its potential for food security and income generation. Smallholders in West Arsi zone of the Rift Valley region are the main source of potato supply to the capital, Addis Ababa, and other major cities in Ethiopia. The Shashemene spot market, in the study area, is considered the main trade hub for potatoes in Ethiopia (Emana and Nigussie, 2011). In this region, potato is the most important single crop for the majority of smallholders. For instance, in the 2009/2010 production season, potato sales accounted for 49% of the total cash income for the sampled households.

In the case examined, farmers can sell their potatoes in nearby markets or sell at farm gate. By selling in nearby markets, farmers could receive the market price less transportation and handling costs. However, due to its bulkiness and perishability, the large majority of farmers opt to sell their potatoes at farm gate.[3] Those

[3] Potato appeared to be the most important farm product to be traded at farm gate, compared for instance to wheat and maize. We think that the bulkiness and perishability of the harvested product are the main explanations. Most farmers do not have good storage facilities and do not have appropriate transport equipment. So farmers depend on middlemen and wholesalers to come to the farm immediately after harvest and take the potatoes with them.

farmers selling at farm gate have two options – to sell directly to wholesalers or to sell through middlemen who act as commissioners on behalf of wholesalers. Wholesalers bring potatoes to the capital city, Addis Ababa, to other major cities in the country and/or to neighboring countries. The same wholesaler may both buy directly from farmers and buy through middlemen.

The study area is characterized by a high involvement of middlemen in potato trading. However, no official data are available about the number of middlemen in the region. Based on primary data we collected from 107 middlemen in the study area in 2011, the average age of the middlemen was 27 years, with an education level of junior high school or less. Of the sampled middlemen, 44% were also farmers, 41% were school dropouts, and 15% were ex-soldiers. During the 2010/2011 production season, one middleman intermediated, on average, between 79 farmers and 7 wholesalers. In working with potato farmers and wholesalers, middlemen can perform several activities, such as matching of buyer and seller demands, negotiating farm gate prices, facilitating harvesting, and organizing transport of potatoes from individual farms to the premises of the wholesaler. Middlemen can also reduce information asymmetry between wholesalers and farmers by monitoring farm activities and providing (to the wholesaler) information on the quality and quantity of potatoes grown. The sampled middlemen received commission only from the wholesalers.

3.1.2. Data

Data were collected in the Spring of 2011 among potato farmers in the Ethiopian Rift Valley Region. A total of 350 farmers participated in the survey. The selection of the samples was carried out as follows. First, we purposely selected the West Arsi zone and three districts and 16 peasant associations.[4] Based on data obtained from the zonal Agriculture and Rural Development Office, the districts Shashemene, Shiraro and Shala have the highest number of potato growers and highest level of production, in a descending order. Then, using a proportional random sampling technique, we selected 63%, 26% and 11% of the sampled farmers from Shashemene, Shiraro and Shala districts respectively. Potato farmers from each peasant association were randomly selected (using systematic random sampling) from the land ownership register that was available in each of the district Office of Agriculture and Rural Development. In the event of unavailability of a farmer, the next farmer in the register was included in the survey. After testing the questionnaire, the survey was personally administered to 350 farmers using five trained enumerators under the supervision of the first author. Data from five potato farmers were dropped during data cleaning for incomplete information.

3.2. Empirical model for determinants of selling through middlemen

As argued above, farmers' choice to trade through middlemen or not could be influenced by socioeconomic and social network related factors. Regarding the latter, individuals may endogenously choose whether to form (or not) a social network with other agents (Bala and Goyal, 2000; Goyal and Vega-Redondo, 2005). Even if the formation of a social network structure is considered exogenous (i.e., caused by socioeconomic conditions external to the community), the decision to maintain it (and at what level of activity) remains endogenous (Krishnan and Sciubba, 2009). Thus, we conceptualize the choice of a trading partner as a standard adoption decision problem (Feder et al., 1985), where the decision to sell through a middlemen or directly to a wholesaler is determined by socioeconomic and social network related variables.

For finding the determinants of the farmer's choice, we estimated a probit model. A probit model seems most appropriate, since the actual level of utility for each farmer U_i is not observed, the part of each farmer's utility function that is observable can be expressed as a function of a vector of exogenous variables X_i and a vector of parameters β to be estimated:

$$V_i(\beta' X_i), \quad where \;\; U_i = V_i(\beta' X_i) + u_i \tag{1}$$

X_i represents a vector of socioeconomic characteristics and variables related to social network structure; and u_i is the unobservable portion of the farmer's utility, which is assumed to be independently and identically distributed.

A farmer will choose to sell potatoes through middlemen if the utility[5] gained from intermediation, U_i^M, is greater than the utility from selling directly to a wholesaler, U_i^D. The probability of a farmer selling his potatoes through a middleman is given by $p(u_i < \beta' X_i)$. The fact that the error term is modeled to have a standard normal distribution motivates the use of a probit model (Wollni and Zeller, 2007). Thus, the model to be estimated is given by:

$$p(M_i = 1) = p(u_i < \beta' X_i) = \beta' X_i + u_i, \quad \text{for } i = 1, \ldots, N, \tag{2}$$

where $M_i = 1$ if $U_i^M > U_i^D$, and $M_i = 0$ if $U_i^M \leq U_i^D$

4. Results and discussion

The first part of this section provides definitions and summary statistics of the socioeconomic and social network variables included in our analysis. Next, we show the correlation between these variables and smallholders' choice of a trading partner and discuss the most important determinants of trading via middlemen. The final part of this section provides a detailed gross profit analysis, based on a sub sample of 110 farmers, and discusses the effect of using middlemen in trade on smallholders' welfare.

4.1. Descriptive statistics

Table 1 provides summary statistics. Of the randomly selected 345 farmers, 68% sold their potatoes through middlemen in the 2009/10 production season. This indicates the dominant role of middlemen in the potato market. The descriptive statistics generally show the presence of significant differences between farmers who did and did not sell potatoes via middlemen such as lagged potato land size (2005/6), total land size in 2009/10 season, total livestock units, revenue from potato sales, income from selling other crops, income from other sources, social network variables, ethnic and religious ties, and location variables, whether a farmer was from Shashemene (thus, near the market) or from the distant districts Shiraro and Shala.

The group of farmers selling potatoes directly to wholesalers appear to be more heterogeneous than the group selling through middlemen, particularly in terms of total livestock units, income from selling other crops and income from other sources. Gender was also one of the network variables which we considered important for this study. However, of the total 350 households who participated in the survey, only 3.5% of them were female-headed households. As the sample size was insufficient to provide any meaningful information, gender was not considered further.

[4] A peasant association covers around 800 ha of land (Hagos et al., 2012), and is the lowest administrative unit in Ethiopia.

[5] Here the farmer's utility may not be limited to economic benefits.

Table 2
Determinants of farmers' choice of trading through middlemen.

Independent variables	*P* (selling through middlemen) (Robust st. err.)
Age of the respondent (in years)	0.026000(0.0127)**
Family size	0.01380(0.0245)
Education in school years (head)	0.101900(0.0321)***
Total livestock units	−0.06600(0.0165)***
Presence of a mobile phone	0.16240(0.2062)
Presence of a cart	−0.05980(0.2026)
Land size (potato) for 2005/6 season	−0.27800(0.1334)**
Membership in a cooperative	0.02010(0.1755)
Household head has ethnic ties with the main middleman or buyer	0.90420(0.2194)***
Household head has religious ties with the main middleman or buyer	0.68140(0.2249)***
Income from sale of other crops, 2009/10 season	−0.00001(0.00001)
Income from other sources including from sale of cattle, 2009/10 season	−0.00001(0.00001)
Farmer is from Shashemene district	0.0606(0.2978)
Farmer is from Shiraro district	0.6516(0.3201)**
_cons	−1.31410(0.5397)**
Wald chi2(12)	100.80
Observations	345
(Pseudo) R^2	0.3440

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

4.2. Determinants of trading with middlemen

Before we present the results, it is important to note how farmers were treated into the middlemen and wholesaler groups. Farmers were grouped into the middlemen and wholesaler channel depending on the amount of potatoes sold (more than 50%) either via middlemen or directly to wholesalers (Appendix A). In both groups, 96% of farmers reported to have used only one channel for selling all of their potatoes. Thus, we did not carry out a separate analysis for those farmers who sold potatoes partly via middlemen and partly directly to wholesalers.

We conceptualize the institution of middlemen as an economic and a social network structure where both social and economic variables are likely to affect farmers' choice. Table 2 displays the observable characteristics used to analyze the determinants of choosing a particular trading partner. Before discussing the results, it is important to discuss potential concerns of endogeneity[6] in the probit model. One potentially endogenous variable is production volume. It is likely that a farmer's decision to trade through a middleman or sell directly to a wholesaler depends on the volume of production (the latter was approximated by land allocated to growing potatoes). Following Rivers and Vuong (1988) and Hernández et al. (2007), we used the area of potato land in the 2005/2006 production season (a lagged variable) as a proxy for volume to avoid potential endogeneity of the production volume[7] in the current period (2009/10). The variable 'membership in a cooperative' is less likely to be endogenous as cooperatives in the study areas did not provide market outlets for potatoes. In the remaining variables, we have no reason to suspect endogeneity.

Results show that older farmers choose middlemen as trading partners rather than selling directly to wholesalers. We would have expected more experienced farmers to be self-reliant in dealing with the market and thus be less dependent on middlemen. From a social network perspective, however, this result is not surprising as older farmers are likely to be more embedded in social networks and have more personalized relationships than the younger ones. This is also consistent with the findings of Koutsou et al. (2014) that younger farmers generally show limited trust in persons and institutions than older farmers. Economic theory would assume that educated farmers have better market information; thus we expect a negative relationship between the propensity to rely on middlemen and farmers' level of education. Results, however, show the opposite and no immediate explanation could be found for this relationship. Farmers with a high number of livestock units are less likely to trade via middlemen, which is against our expectation. One possible explanation is that having more units of livestock could give farmers insurance against price risks and encourage them to directly trade with wholesalers. Furthermore, the opportunity cost of labor may not be as high as we expected due to the large size of the families in the study areas. Land size, as a proxy for volume, was an important determinant. The results show that farmers who grow potatoes on a relatively large scale were less likely to choose a middleman as their trading partner. We would have expected otherwise because a large production means high market risk (particularly the risk of not finding a buyer for all products) and thus leads farmers to use middlemen to search for an appropriate buyer. However, economies of scale appeared to be more important than price risk. Having a large area of potato is more likely to attract wholesalers, who seek to minimize their transaction cost. The presence of a cart in the household is negatively correlated to the use of middlemen, as owning such asset might help farmers to transport potatoes to nearby markets (in case no wholesaler appears at the farm gate). The main human asset variable - family size - did not affect the choice of a trading partner mainly because there was little variability between the two groups of farmers. We also controlled for income generated from the sale of other crops and from other sources such as the sale of livestock and remittances, but we did not find significant differences between the two groups.

As expected, the social network variables, ethnic and religious ties, showed a significant and positive relationship in choosing middlemen as trading partners. The probability of using middlemen increases when a farmer has ethnic and/or religious ties with a middleman. From the location dummy variables, Shiraro did show a significant and positive relationship with the use of

[6] The problem of endogeneity arises when the hypothesized relationship between the independent variable and the dependent variable is systematically correlated with the error term (due to other unobserved factors), which causes bias in the parameter estimates.

[7] To overcome potential endogeneity concern, we took the 2005/2006 land area rather than of the 2009/2010 season in the probit model. For example, parameter estimate for potato land could be biased if farmers' prior knowledge about the merits of trading via middlemen had influenced the current land allocated for growing potatoes. Based on our analysis, farmers in the sample had on average a 3-year long relationship with the main trading partner. This means that the 2005/2006 potato land could be used as a proxy to measure the portion of land allotted by farmers for growing potatoes a year before they entered into trading relationships with the current main buyer. For this purpose farmers were retroactively asked how much land was used to grow potatoes for the seasons from 2005/2006 to 2009/2010 measured in the local unit "timad", which is equivalent to 0.25 ha.

middlemen. This was expected; compared to the reference district Shala, which is located between Shashemene and Shiraro. Farmers from Shiraro are more likely to use middlemen in potato trading because they are at a larger distance from Shashemene, the main hub of potato trading in the region. Cooperative membership did not have a significant effect on a farmer's choice of a trading partner, while we expected members to be less dependent compared to non-members. An explanation from earlier findings is that cooperatives in Ethiopia are largely ineffective in improving smallholder commercialization[8] (Bernard et al., 2008; Ruben and Heras, 2012). We found that over 50% of farmers from both groups had access to inputs via cooperative. However, this did not have a significant effect on the decision to trade via middlemen, mainly due to the low variability between the two groups of farmers. Also, having a mobile phone did not have a significant effect on using middlemen as trading partners. The majority of farmers in both groups had mobile phones, and there was little variability between the two groups.

4.3. The effect of trading with middlemen on smallholders' welfare

The question we seek to answer here is that whether trading via middlemen affects smallholders' welfare. We carried out a gross profit analysis for farmers who sold their potatoes through middlemen versus those who sold directly to wholesalers.[9] In the survey, every other respondent from the randomly drawn list was asked detailed information needed for the gross profit analysis. After excluding five respondents for incomplete information, 145 farmers were considered for the following analysis. We took a cautious approach to analyzing gross profit by including only those farmers who sold their potatoes at the farm gate, thus excluding those farmers incurring transport and marketing costs by selling in nearly markets. Of the 145 farmers in the sub sample, 110 had sold their potatoes at farm gate, of which 30% sold directly to wholesalers and 70% sold through middlemen. This representation of the two groups of farmers in the sub sample closely matched with that of the larger survey, which had 32% and 68% respectively.

Table 3 provides a direct comparison of the amount and cost of inputs used during production as well as part of marketing costs, realized revenue and gross profits, and revenue and gross profit based on the value of production per hectare. Both groups of farmers appeared to be heterogeneous in terms of cost, yield, revenue and gross profit. Particularly, standard deviations for the realized revenue and realized gross profit were higher than the standard deviations reported based on the value of production. This was mainly because farmers who had not sold their potatoes at the date of the interview were recorded at 'zero' for the purpose of reporting the realized revenue and gross profit, and at 'market value' for the purpose of reporting the revenue and gross profit based on value of production.

Following Neven et al. (2009), wages paid for harvesting activities were itemized as a marketing expense. However, marketing expenses are not limited to harvesting activities. Particularly for bulky and perishable agricultural crops like potatoes, the cost of transportation is substantial and more likely to vary depending on the location of the farm. Our conservative approach to focus on farmers who sold at farm gate is expected to generate a more robust result.

4.3.1. Price and yield differences

As shown in Table 3, a significant difference in price and yield was observed between the two groups of farmers. First, farmers selling directly to wholesalers used more or less similar levels of inputs (seed, chemical and fertilizer) but had some difference in cost per unit of inputs per hectare. Farmers in the study area often buy chemical and fertilizer from cooperatives which do not require membership to buy inputs. Thus, the unit cost would not change between the two groups of farmers. However, potato seeds were not provided by cooperatives, which might explain the increased cost of potato seeds (8.6%) paid by farmers who sold directly to wholesalers. The wage rate paid by farmers selling directly to wholesalers was higher than the rate paid by the 'middlemen' group. In sum, the total cost per hectare paid for inputs by the 'wholesaler' group was relatively higher than those farmers who traded via middlemen, at 10% significance level. This may suggest a difference in the quality of inputs used and consequently a difference in yield between the two groups of farmers, which was 32.7%. Second, there was a substantial difference in the gross profit per hectare between the two groups of farmers. Compared to those farmers who sold through middlemen, the gross profit realized per hectare was 7500 Birr for farmers who sold directly to wholesalers versus 2307 Birr for farmers who chose middlemen as trading partners. This substantial difference in gross profit could be explained by the differences in yield and price, both of which were found to be statistically significant. The yield gap could be linked to the quality of inputs used in production. As noted above, those farmers who opted to sell through middlemen were relatively low-resource endowed and might not have had enough money to pay for the more productive labor force and improved seeds. However, the price difference is important to note. Though market-oriented farmers in the region generally grow the 'Nech Abeba' variety, which is also considered the main commercial potato variety (Abebe et al., 2013), the average price received by smallholders trading through middlemen was 15% lower than the price received by farmers selling directly to wholesalers. Given our cautious approach to consider only farmers who sold at farm gate, one would expect similar prices to be received by the two groups of farmers.

Our explanations for the price difference are the following. Middlemen are embedded in the community, where receiving money from fellow farmers in the form of 'commission' is considered an unacceptable practice. Farmers who used middlemen for selling their potatoes mentioned that they did not pay any commission. The amount of commission paid by traders ranged between 3 and 12 Birr, with a mean and standard deviation of 6.3 and 2.1 respectively, per 100 kg potato sales. We speculate that this difference in the level of commission paid by traders might be related to the price that the middleman negotiates with the farmer. We expect that a lower price paid to the farmer, which thus represents a lower procurement cost for the wholesaler, will lead to a higher commission for the middleman. From literature we know that middlemen generally have a strong bargaining position due to their knowledge of supply and demand conditions at the farm and in the market (Neven et al., 2009). Another explanation might be that, with intermediation, the marketing channel becomes longer and thus a smaller percentage of the marketing margin can be captured by the farmer. A third explanation for the price difference could relate to potential differences in contract specifications. Smallholders may be locked-in pre-agreed prices and credit advances to buy inputs such as seed and fertilizer. Such arrangements could lead to lower margins for farmers. To see whether this was the case we have analyzed the presence of any form of contract and the specifications governing relationships between farmers and traders. Farmers reported that no formal contracts were used in the

[8] Generally, the term commercialization is defined as the volume of produce and household resources that enter the exchange economy, which can occur on the output side of production with increased marketed surplus as well as on the input side with increased use of purchased inputs (Von Braun, 1995).

[9] Data related to farmers selling directly to wholesalers also included farmers selling to anonymous buyers, which accounted for only 10% of the total sample size.

Table 3
Gross profit analysis of potato farmers selling through middlemen versus directly to wholesalers.

Items (per ha)	Unit	Farmers selling through middlemen (n = 77)				Farmers selling directly to wholesalers (n = 33)			
		Qty	Unit cost	TC	%Rev	Qty	Unit cost	TC	%Rev
Seed	100 kg	16	175	2856	24%	17	190	3288	18%
Inorganic fertilizer	kg	249	8	1938	16%	280	8	2228	12%
Chemicals	L	34	9	309	3%	34	9	287	2%
Oxen rental	ha	1	323	323	3%	1	324	324	2%
Land rental	ha	1	564	564	5%	1	615	615	3%
*Total input cost**				*5991*	*50%*			*6743*	*37%*
Labor cost by activity									
Land preparation	Mdys[a]	8	75	595	5%	8	88	667	4%
Planting	Mdys	14	75	1062	9%	15	88	1292	7%
Weeding	Mdys	11	75	797	7%	11	88	1010	6%
				2454	*20%*			*2969*	*16%*
Labor cost by source									
Family	Mdys	22	75	1626	14%	24	88	2074	11%
Hired	Mdys	11	75	828	7%	10	88	895	5%
*Total production cost**				*8445*	*70%*			*9712*	*54%*
Marketing									
Harvesting	Mdys	16.4	75	1233	10%	15	88	1361	8%
Harvesting by labor sources									
Family	Mdys	4.5	75	335.9	3%	5.5	88	488	3%
Hired	Mdys	11.9	75	896.9	7%	9.9	88	873	5%
Total marketing cost				*1233*	*10%*			*873*	*5%*
Total costs				9678(7981)	81%			10,585(10888)	59%
Realized revenue***				11,985(9594)	100%			18,085(13717)	100%
Gross profit-realized**				2307(4537)	19%			7500(8773)	41%
Revenue, value of production per ha				15,431(10667)				23,035(15547)	
Gross profit, value of production**				5753(3914)				12,450(7205)	
Yield**				107(69)				142(101)	
Commercialization Index				0.75(0.25)				0.69(0.29)	
Unit price*				150(58)				172(71)	

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$. [a]Man-days. 1US$ was approximately equal to 12.60 Birr in 2009/2010 season.
Note: Figures in brackets represent standard deviation of mean values.
Items in italics are section summary totals.

trading relationship. Nevertheless, 97% of farmers selling via middlemen and 70% of farmers selling directly to wholesalers reported to have had some form of prior agreement, particularly about prices (Table 4). Specifications related to credit advances or provision of seed, fertilizer and other inputs were rarely included in such agreements. The fact that almost all farmers who used middlemen had a pre-agreed price partly justifies the lower gross profit margin for this group in Table 3.

As a robustness check, we also estimated the impact of smallholders' use of middlemen on farmers' income related to potato sales using propensity score matching (PSM), a method which helps to address the problem of self-selection bias (Heckman et al., 1997). This bias can occur in the case examined because one cannot observe how farmers' income from potato sales would have looked like without using middlemen. Therefore, we estimated the average treatment effect on the treated (ATT). Although other methods are available to estimate ATT, we chose the radius matching (RM) estimator because it uses not only the nearest neighbor within each caliper but also all of the comparison members within the caliper (Dehejia and Wahba, 2002) and allows for use of extra (fewer) units when good matches are (not) available (Khandker et al., 2010), and thus avoids the risk of bad matches in case the closest neighbor is far away (Caliendo and Kopeinig, 2008). After controlling for selection bias using PSM and checking for the matching quality (Appendices B and C), the difference in revenue from potato sales between farmers who sold directly to wholesalers and those who sold via middlemen was Birr 5,940, with mean standard error of 3313 (Appendix D); this difference was significant at 10% level.[10] This reinforces the results of the gross profit analysis in Table 3 that selling via middlemen may reduce smallholders' welfare.

4.3.2. Why do smallholders sell through middlemen?

Despite receiving lower prices, the majority of farmers continued to trade through middlemen (for at least the three years leading up to the date of the survey). We used socioeconomic and social network theories to explain why farmers and wholesalers favor trading through middlemen. First, intermediation may be an attractive option for wholesalers because buying from farmers directly may not guarantee sufficient supply in a region where trading relations are socially tied. Thus, trading through middlemen would give buyers a competitive advantage.

Second, the presence of any form of quality uncertainty in trade may encourage both wholesalers and farmers to use middlemen. Based on our own observations in the field and own knowledge of the market conditions in Ethiopia and on interviews with a few traders during the study period, we think that quality uncertainty provides an important explanation. For example, supplying premature tubers can easily lead to damaged skins during product handling, transportation and storage. This means that suppliers of good and bad quality potatoes need to be screened or that additional costs have to be incurred for measuring quality at the time of purchase. Middlemen, living in the community of the farmers, have much lower cost in monitoring farmers and/or measuring quality than wholesalers who are from outside the community. As smallholders produce and supply potatoes in small quantities, wholesalers would have to spend much time inspecting the potatoes delivered by each farmer. Rather, wholesalers do random checks to

[10] However, the results of the PSM can only be used as a robustness check as the main outcome variable used to measure the difference in welfare between the two groups was revenue (cash received from potato sales) in 2009/2010. The analysis was based on data from the 345 farmers but did not take into account potential differences in production and marketing costs. Detailed cost data were available only for the 145 farmers included in the gross profit analysis in Table 3.

Table 4
Contractual relationships between farmers and wholesalers in the potato trade.

Description	Farmers sold via middlemen	Farmers sold directly to wholesalers
Contract forms (%)		
• Prior written/formal contracts	0	0
• Some form of prior informal/oral agreements	97.4	70
• No form of prior agreements	2.6	30
Contract specifications		
Price (%)		
• Pre-agreed price	96.6	68.2
• No pre-agreed price	3.4	31.8
Fertilizer (%)		
• Provision of fertilizer by buyer	2.1	0
• No provision of fertilizer by buyer	97.9	100
Seed (%)		
• Provision of seed by buyer	0.4	0
• No provision of seed by buyer	99.6	100
Credit (%)		
• Provision of credit by buyer	2.1	1.8
• No provision of credit by buyer	97.9	98.2

verify quality. Farmers usually supply potatoes in sacks in which only the top part is visible for inspection, and random quality checking is done for the top part. Wholesalers just assume the bottom part of the sack to contain poor quality potatoes. Subsequently, wholesalers tend to pay an average price that may not reflect the actual quality of potatoes. This practice discourages farmers who deliver good quality potatoes, which can affect future relations. Therefore, the presence of a middleman can resolve this adverse selection problem by taking care of the interests of both wholesalers and farmers. In fact, middlemen reduce the measurement cost problem (Barzel, 1982) as they regularly visit the farm during cultivation and harvest.

The third explanation why farmers and wholesalers favor middlemen relates to commercialization. Those farmers who participated in the middlemen channel were the low-resource endowed ones. They had, on average, smaller farm sizes, a lower number of livestock units and lower income from selling other crops. These farmers are more likely to be excluded from the relatively high value market channels as transaction cost per farmer would be too high for wholesalers to buy directly from farmers. Trading via middlemen can thus be a viable strategy for smallholders to be linked to wholesalers and final markets. In the case examined, farmers' choice for middlemen as trading partners had enhanced smallholder commercialization. Farmers who participated in the middlemen channel were able to turn their stock of potatoes into cash better than those farmers who sold directly to wholesalers; the commercialization index[11] for the former was 0.75 versus 0.69 for the latter.

The fourth reason relates to the effect of social networks on the trading relationship. As shown in Tables 1 and 2, it appears that social ties are weak between farmers and wholesalers who directly buy from these farmers. On the other hand, the large majority of farmers participating in the middlemen channel had strong ethnic and/or religious ties with those middlemen. Such ties could lead individuals to cooperate according to social norms (Fehr et al., 1997; Hoffman et al., 1998) even though such cooperation would lead to negative economic outcomes (Ali and Peerlings, 2011). In light of this, Granovetter (1985) argues that socially tied relations may lay a necessary condition for trust and trustworthy behavior; however, such trust may present enhanced opportunities for 'malfeasance'. Social networks may also restrict business exchange to a limited number of agents who can then manipulate the exchange process by controlling information on prices and markets (Alesina and La Ferrara, 2005). The continued use of middlemen by smallholder farmers, despite the lower profit margin, could also relate to the theory on "economy of affection". According to this theory, social ties such as religion and ethnicity can hold individuals and communities together in trade relationships (Hydén, 1983).

As a final note, although attempts were made to minimize potential problems of endogeneity, some variables like 'total livestock units', 'other income sources' and 'income from sales of other crops' could be endogenous in our probit model. Also, other unobservable covariates may affect the results of our study. For instance, farmers' choice of a trading partner and profitability may still be affected by entrepreneurship traits which we attempted to capture by using age and education level. Thus, the parameter estimates may not necessarily reflect causal relationships and may thus require careful interpretations.

5. Conclusion

Whether middlemen have a positive or negative impact on smallholder commercialization and smallholder welfare is an unresolved issue in the development literature. Middlemen may solve market imperfections which are ubiquitous in developing countries (e.g. Maertens and Barrett, 2013), but they may also take a relatively large share of the profit margin in the value chain and they may constrain upgrading opportunities for smallholder farmers (e.g. Lee et al., 2012). This paper has analyzed the factors that determine the farmers' decision to trade through middlemen and the effect of that decision on their welfare. For this purpose, we conceptualized the institution of middlemen as an economic institution and as a social network structure embedded in a community. The social ties in the social network structure facilitate economic exchange between farmers and middlemen and thus also between the farmers and the wholesalers for which the middlemen work. While earlier studies focused on the social network structure of traders themselves (e.g. Fafchamps and Minten, 1999), our study analyzed specifically the social networks to which both farmers and middlemen belong.

We found that ethnic and religious ties between farmers and middlemen – our main social network variables – are important factors in shaping exchange relationships in the context of potato trading in the Rift Valley Region of Ethiopia. Most importantly, our paper provides answers as to why smallholders may still continue trading via middlemen even when this choice of sales arrangement might reduce welfare. While trading via middlemen appeared to enhance smallholder commercialization, selling directly to wholesalers seemed more appropriate in enhancing the welfare of the

[11] In this paper, the commercialization index is defined as the ratio of the actual potato sales to the total value of production by a household during the study period.

better-resource endowed farmers. In general, consistent with some previous studies, such as de Weerdt (2002) and Munshi (2014), our findings assert that religious and ethnic ties provide a foundation for socioeconomic exchanges in rural economies. However, the effect of social networks on smallholders' welfare is mixed. This means that it would be difficult to generalize social networks as having a net loss or gain in welfare. For example, Ali and Peerlings (2011) found a net negative effect of ethnically tied trade on the welfare of small-scale producers in the handloom sector of Ethiopia. In contrast, Gabre-Madhin (2001) reported significant welfare gains among grain traders who used the institution of middlemen in trade. These findings seem to suggest that more studies are needed to better understand context specific social network variables affecting socioeconomic outcomes in rural economies.

There are several suggestions to be made for further research. First, our human capital variables, age and education, show a strong positive relationship with the use of middlemen in trade although we expected that more experienced and educated farmers would choose to sell directly to wholesalers because these farmers have a better capacity to deal with the requirements of the wholesalers. Part of our explanation is that older farmers might have stronger ties with middlemen, and the importance they give to these ties makes them continue to work with the middlemen. We believe future research should explore the relationship between human capital and socially-tied relationships in trade and the effect on smallholder welfare. Second, resource endowments such as having many livestock units, having plenty of family labor available and owning a cart did not affect the choice of trading partner. We think that the low opportunity cost of labor in rural communities makes these resources less important in determining the choice of a trading partner. Future studies should provide more evidence on this relationship. A third suggestion for further study relates to measuring transaction costs. Although our study highlighted how middlemen can optimize monitoring and measurement costs, no attempt was made to actually measure transaction costs, such as search and negotiation time, contract enforcement and commitment failure costs. Indeed, measuring transaction costs is a difficult task as most costs fall in the category of potential costs, or risks. While very few studies have actually measured transaction costs, progress in developing measuring methods has been made (Benham and Benham, 2005). Future research should therefore try to measure the transaction costs faced by farmers using different sales arrangements. Fourth, we only included two network variables in our probit model. One network variable we did not include in further analysis was gender, because of insufficient data. Yet, gender is a significant factor in social networks linked to rural farming (Anthopoulou, 2010). We thus encourage similar studies to ensure the representation of female-headed households in the sample. Also, it would be interesting to know how farmer participation in other social networks, such as 'iddir', 'debbo', and 'equb' in Ethiopia, could affect trading relationships. Furthermore, we have not attempted to measure any potential social switching costs that may arise if a farmer decides to sell directly to a wholesaler instead of using a middleman. Further research may go this direction to expand our understanding of the effect of social networks on trade.

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Appendix A

An overview of farmers' choice of a trading partner in selling potatoes in the 2009/2010 production season.

Farmers who sold:	Frequency	Percent
More than 50% via middlemen	235	68.1
More than 50% directly to wholesalers	75	21.7
More than 50% directly to anonymous buyers	35	10.1
Partly via middlemen (≥50%) and partly to others (<50%)	10	2.9
Partly to wholesalers (≥50%) and partly to others (<50%)	5	1.4
Partly to anonymous buyers (≥50%) and partly to others (<50%)	7	2.0
100% via middlemen	225	65.2
100% directly to wholesalers	70	20.3
100% directly to anonymous buyers	28	8.1

Appendix B

Test of matching quality (radius matching, caliper = 0.05).

Variable	Treated	Control	% Bias	p-Value after matching
Age of household head (in years)	36.02	37.36	−12.8	0.25
Family size	9.38	9.34	0.8	0.94
Education (in school years)	5.84	5.75	2.6	0.81
Total livestock units	5.33	5.65	−1.5	0.51
Presence of a mobile phone	0.7	0.71	−3.1	0.75
Presence of a cart	0.4	0.44	−7.7	0.44
Potato cultivated area in 2005/6 (ha)	1.06	1.11	−4	0.58
Membership in a Cooperative	0.58	0.62	−7.9	0.42
Household head has ethnic ties with middlemen or buyer	0.7	0.69	2.2	0.83
Household head has religious ties with middlemen or buyer	0.62	0.62	1.1	0.92
Farmer is from Shashemene district	0.67	0.66	2.4	0.81
Farmer is from Shiraro district	0.2	0.22	−4.4	0.67

Appendix C

Matching quality indicators before and after matching (radius matching).

Matching estimator	Pseudo-R2		LR^2 of probit model		Observation retained	Mean standardized bias		Total % |bias| reduction
	Before	After	Before	After		Before	After	
Radius matching								
Caliper = 0.05	0.34	0.01	101 (p = 00)	3.7(p = 0.97)	316	21.3	4.2	80
Caliper = 0.03	0.34	0.01	101 (p = 00)	5.7(p = 0.93)	316	23.7	5.2	78

Notes: Estimations for the propensity score, covariate imbalance testing and the common support graphing for the different types of matching estimators were performed by using the program psmatch2 provided by stata 12.0 software.

Appendix D

Average treatment effect for the treated based on Radius matching.

Variable	Farmers selling through middlemen	Farmers directly sold to wholesalers	ATT	SE[a]
Potato sales in 2009/2010 (in Birr[b])				
Caliper = 0.05	7732	13,672	−5940*	3313
Caliper = 0.03	7732	13,243.	−5511*	3151

[a] Standard errors are bootstrapped with 500 replications. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.
[b] 1US$ was approximately equal to 12.60 Birr in 2009/2010.

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